Loan Rangers

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Problem Space

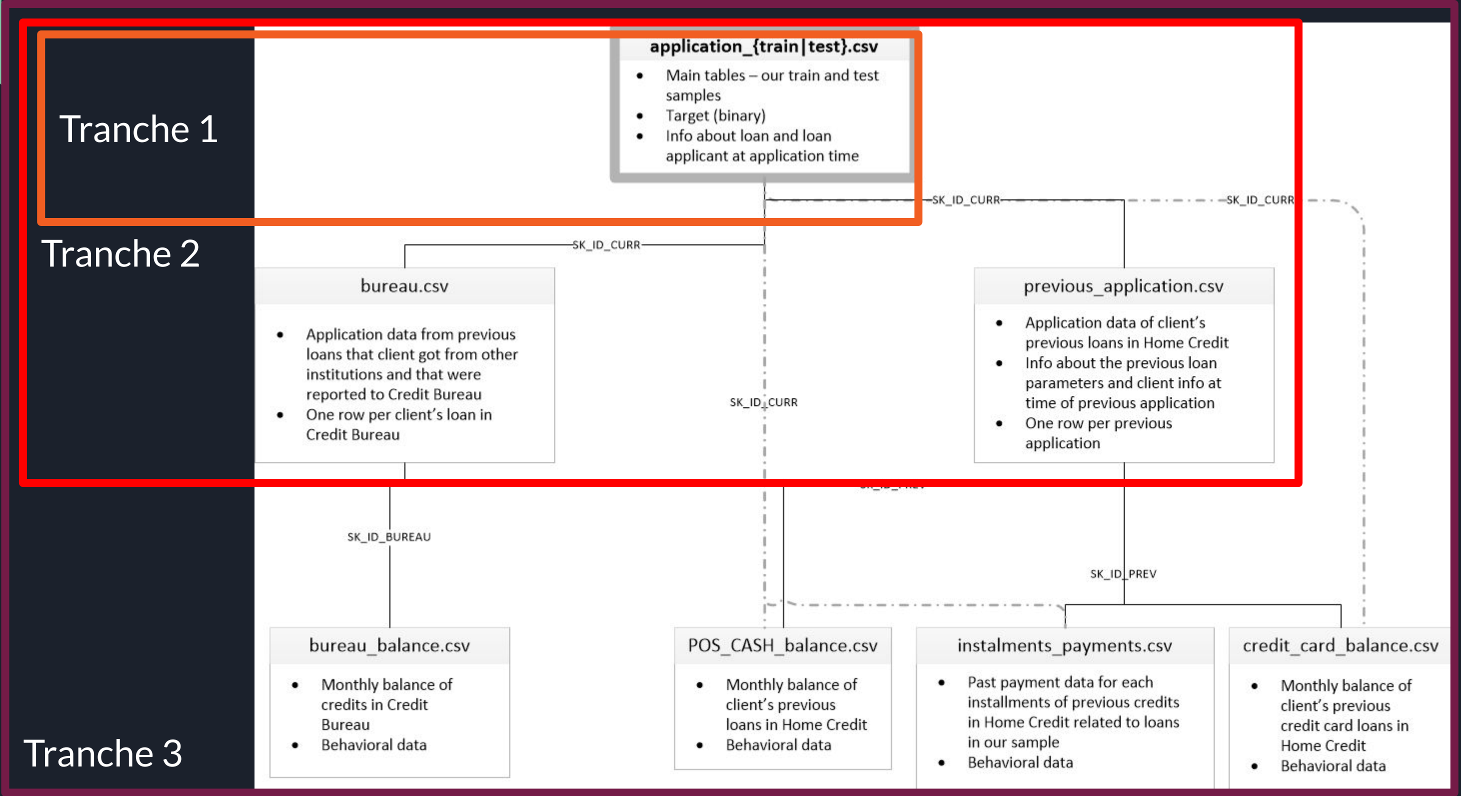
kaggle

HOME CREDIT GROUP · FEATURED PREDICTION COMPETITION · 6 YEARS AGO

Home Credit Default Risk

Can you predict how capable each applicant is of repaying a loan?

Dataset



Tranche 1
Tranche 2
Tranche 3
application_{train|test}.csv
Main tables – our train and test samples
Target (binary)
Info about loan and loan applicant at application time
SK_ID_CURR
SK_ID_CURR
SK_ID_CURR
SK_ID_CURR
bureau.csv
Application data from previous loans that client got from other institutions and that were reported to Credit Bureau
One row per client's loan in Credit Bureau
previous_application.csv
Application data of client's previous loans in Home Credit
Info about the previous loan parameters and client info at time of previous application
One row per previous application
SK_ID_BUREAU
SK_ID_PREV
bureau_balance.csv
Monthly balance of credits in Credit Bureau
Behavioral data
POS_CASH_balance.csv
Monthly balance of client's previous loans in Home Credit
Behavioral data
instalments_payments.csv
Past payment data for each installments of previous credits in Home Credit related to loans in our sample
Behavioral data
credit_card_balance.csv
Monthly balance of client's previous credit card loans in Home Credit
Behavioral data

Cleaning, Encoding, Selecting and Resampling Data

Data Cleaning: Removed columns and rows with NaNs that cannot be otherwise encoded suitably (Encoded as ‘missing’ in categorical data or the maximum distance from other data if continuous)

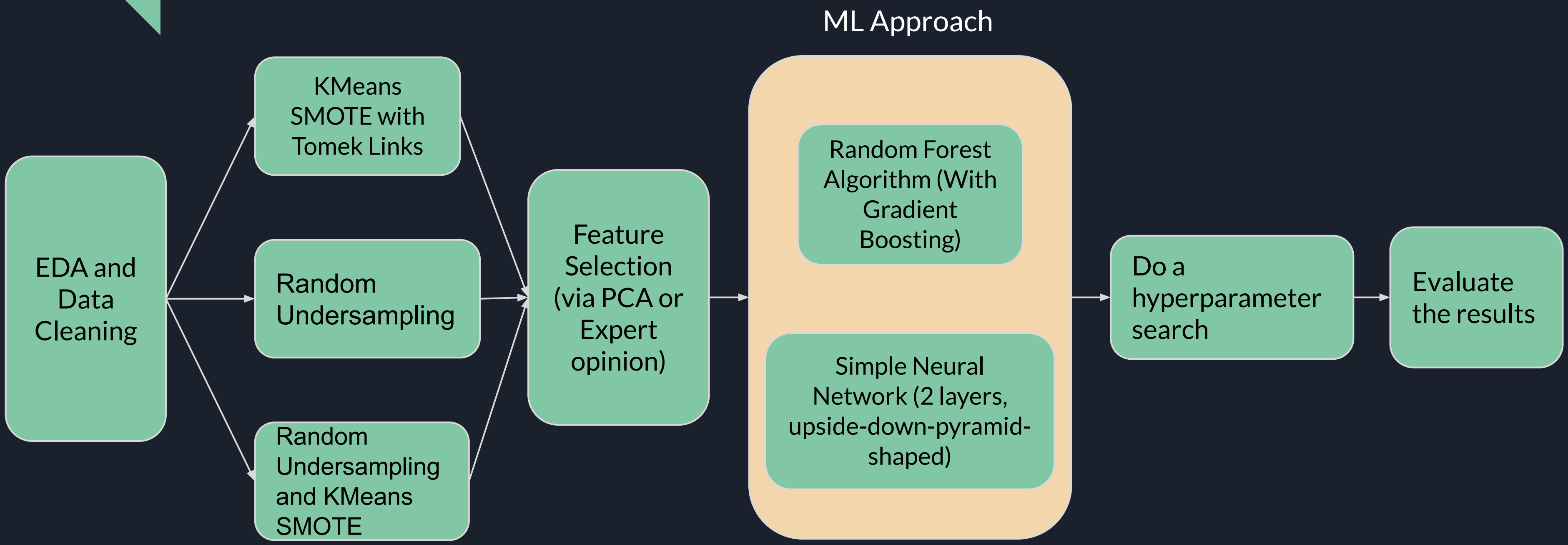
Encoding Data: Performed 1-hot encoding to turn all data into numerical values.

Encoding Data from Secondary Tables: Computed the mean, median and variance of the samples mapped to the main dataset to try and capture the trend from multiple samples.

Selecting Data: Performed a Principal Component Analysis to identify the components that explained the most variance

Resampling Data: Performed Random Undersampling, KMeans Synthetic Minority Oversampling Technique (SMOTE) and a combination of both to rebalance classes

Approach


ML Approach
EDA and Data Cleaning
KMeans SMOTE with Tomek Links
Random Undersampling
Random Undersampling and KMeans SMOTE
Feature Selection (via PCA or Expert opinion)
Random Forest Algorithm (With Gradient Boosting)
Simple Neural Network (2 layers, upside-down-pyramid-shaped)
Do a hyperparameter search
Evaluate the results

Results

1. Random Forest Algorithm (With Gradient Boosting)

	Tranche 1	Tranche 2	Tranche 3
Training	0.65550	0.64966	0.61137
KMeans SMOTE with Tomek Links	0.62566 0.62880	0.62694 0.63794	0.63791
Random Undersampling	0.65537	0.64966	0.61137
Random Undersampling and KMeans SMOTE	0.71708 0.72400	0.67342	0.61361

HyperParameter-Optimized Score is highlighted

2. Simple Neural Network (2 layers, upside-down-pyramid-shaped)

	Tranche 1	Tranche 2	Tranche 3
Training	0.69650	0.69843	0.70503
KMeans SMOTE with Tomek Links	0.65663	0.66357	0.64474
Random Undersampling	0.69232	0.68897	0.70134
Random Undersampling and KMeans SMOTE	0.69404	0.70722	0.70098

Future Extensions

Feature Engineering:

- Embed more useful information using automated feature tools

Efficient Hyperparameter Tuning:

- Leverage Bayesian optimization or evolutionary algorithms

Monetary Risk Integration:

- Incorporate financial metrics (expected returns, losses) for holistic decision-making
- Align predictive risk scores with economic outcomes for more actionable insights